

Head Trauma

History

- Mechanism of injury? Timing of injury?
- Loss of consciousness?
- Other associated injuries, or isolated head injury?
- If applicable, was patient wearing helmet?
- Penetrating injury? (Gunshot wound, stabbing, impaled object)
- Did patient initially have normal mental status, then suddenly deteriorate? (Suggest epidural hematoma)
- Age of patient? (Elderly patients have a higher risk for asymptomatic, early intracranial bleeds with relatively minor mechanism of injury such as a fall)
- Past medical history? Medications? (Is patient on Plavix, aspirin, warfarin, Pradaxa, Eliquis, or other similar drugs?)

MARCHES Protocol

- **Massive Bleeding control** - Tourniquet, Wound packing
- **Airway** – NPA/OPA/Cricothyrotomy
- **Respiratory** – decompress chest if tension pneumothorax suspected, occlusive dressing for open pneumothoraces
- **Circulation**- IV/IO, pelvic binder, Reassess tourniquet/wound packing
- **Hypothermia care**/Hike vs Helicopter (consider evac options)
- **Eye injuries** – cover with rigid shield and no pressure on the eye
- **Spinal Motion Restriction** if indicated

Key Concepts

- DO NOT remove penetrating objects to the skull, neck, or eye but rather splint and support in place
- Minimize scene time and transport to appropriate trauma center
- Aspirin/ Plavix/ Brilinta are irreversible antiplatelet agents—and 'blood-thinners' (like Coumadin, Plavix, Eliquis, Xarelto) can cause serious bleeding from minor trauma

Head Injury Classification

- **Mild** – Low energy injury, no loss of consciousness, may have mild headache and nausea, may be dazed, seeing stars but GCS is 15 (Eg. Hit in head with fists, fall from standing with no LOC)
- **Moderate** – Moderate energy injury, may have loss of consciousness < 5 minutes, confusion, nausea / vomiting, lethargy, dizziness, GCS 13-15 (Eg. MVC, sports injuries, hit with heavy or moderate velocity object)
- **Severe** – Moderate to high energy injury, seizures, loss of consciousness > 5 minutes, altered mental status, any penetrating wound, GCS < 13 (Eg. High speed MVC with altered LOC, high velocity object to head, gunshot wound / stabbing)

EMT

- Stop **Massive Bleeding** as first action, Pressure dressing with firm direct manual pressure
- Establish Simple **Airway** with NPA or OPA and suction secretions if needed
- Support **Respirations**- Apply High Flow Oxygen by Non- Rebreather Mask
 - BVM ventilations for low respiratory rate or End Tidal CO₂ >35 Goal EtCo₂ 40mmHG
- Support **Circulation**
 - Reassess wounds and do Head to Toe Trauma Survey
- Assess **Disability**- AVPU level of consciousness with GCS
- Protect from **Hypothermia**
- **Eye injuries** -cover with rigid shield and no pressure on the eye
- **Spinal motion restriction** if indicated
- Prepare for transport or move patient towards medic unit

Paramedic

- Assess BLS **MARCHES** interventions. High Flow Oxygen to maintain O₂ Sats > 94%
- **Limit scene time** with only EMT level skills or life saving interventions to be done prior to load
- Transport with head elevated 30 degrees if possible
- **Airway**-High Flow oxygen, Capnography is required if physically available - Consider advanced airway if BVM ventilation is not effective
- **Respiration**- BVM ventilation with ETCo₂ Goal 40 mmHg-ventilate to keep between 35-45 mmHg
- **Circulation**- Start IV/IO and administer fluids according to Trauma General Patient Care Guideline
 - If suspected TBI maintain SBP >90 (consider early fluids/blood if BP dropping or elevated HR>110)
- Treat seizures, as per Guideline, and consider sedation if needed for adequate treatment in agitated or violent patients.

DO NOT GIVE TXA TO ISOLATED HEAD TRAUMA